# Siam Rahman Karip

siam.rahman.karip@g.bracu.ac.bd

## About

I am an independent researcher interested in large language models, spanning their evaluation, interpretability, and behavior across languages and settings. I remain open to a range of directions within this space and enjoy discovering new questions as I go.

## Education

- **B.Sc in Computer Science and Engineering** *Feb 2021 - Jan 2025*
  *BRAC University* Dhaka, Bangladesh
  - CGPA: 3.59/4.00 (Distinction)

## Research Interests

Large Language Models (LLMs), Natural Language Processing (NLP), Trustworthy Machine Learning

## Publications

C=Conference, W=Workshop

**[C.1]** Nahid Hossain*, **Siam Rahman Karip***, Md Faisal Kabir, Swakkhar Shatabda and Salekul Islam. 2026. **Systematic Failure Analysis of Open-Weight LLM Annotators for Classification Tasks: Taxonomy, Uncertainty Signals, and Deployment Boundaries**. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, pages 46409–46466, Budapest, Hungary. Association for Computational Linguistics. (*Equal contribution)

**[W.1]** **Siam Rahman Karip** and Nahid Hossain. 2026. **AlienAnnotators at PsyDefDetect: What Lies Between the Lines: Probing Lightweight Open-Source LLMs for Psychological Defense Mechanism Detection**. In *Proceedings of the of the BioNLP 2026 (Shared Tasks)*, pages 213–223, San Diego, California, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2026.bionlp-2.28

## Academic Experience

- **Research Assistant (RA)** *April 2025 – July 2026*
  *United International University* Dhaka, Bangladesh
  **Supervisor:** Mr. Nahid Hossain, *Assistant Professor and UG Program Coordinator*
  - Conducting research in large language models and natural language processing, with a focus on understanding model behavior, evaluation, and reliability.
  - Conducting occasional classes, quizzes, and exam-script grading.
- **Private Tutoring** *July 2022 – May 2025*
  - Tutored university-level students in Python programming, covering core concepts and practical problem-solving.
  - Assisted A-Level students in Pure Mathematics and Statistics, strengthening analytical and quantitative skills.

## Research Projects

- **A Failure Mode Analysis of Autoregressive vs. Diffusion Language Models on Sentence-Pair Classification Tasks** *In Progress* *Target: ACL*
  - Benchmarking autoregressive and diffusion language models zero-shot on different sentence-pair classification datasets under generation and native scoring protocols.
  - Analyzing failure modes beyond accuracy, including per-class behavior, calibration, shortcut sensitivity, and error severity, to test whether different diffusion models fail in the same ways as each other and as autoregressive models.
- **EEG and Eye-Tracking Feature Contextualization based Affective Computing through LightGBM enabled Stacked Ensembling of LLMs** *Undergraduate Thesis* *BRAC University*
  - Introduced specific contextualized formatting for optimal tokenization.
  - Used LLMs (FLAN T5, GPT-2 and BERT) for classifying multimodal physiological numeric data.
  - Merged predictions from multiple LLMs using Majority Voting and Stacked Ensemble Learning Methods.

## English Language Proficiency

**IELTS Academic** *October 2026*

| Listening | Reading | Writing | Speaking | Overall |
|---|---|---|---|---|
| 7 | 7.5 | 7 | 6.5 | 7 |

## Honors and Awards

- **VC's List and Dean's List**
  *BRAC University*
  ◦ Appeared on BRAC University's VC's list and Dean's list for last 3 semesters consecutively.
- **Academic Excellency Award**
  *Uttara North City Corporation*
  ◦ Awarded for achieving a perfect GPA of 5.0 on all subjects in the Secondary School Board examination.

## Courses and Certifications

- [🔗] **Transformer Models with Pytorch** *Datacamp*
- [🔗] **Large Language Models (LLMs) Concepts** *Datacamp*
- [🔗] **AI Fundamentals** *Datacamp*
- [🔗] **Intermediate Python** *Datacamp*

## Skills

- **Programming Languages:** Python, C, Java, SQL, HTML, CSS, LaTeX
- **Tools for Deep Learning:** PyTorch (CUDA), TensorFlow, Transformers (Hugging Face), BitsandBytes
- **LLM Tooling and Evaluation:** Hugging Face Datasets, Hugging Face Evaluate, Scikit-learn
- **Data Science and Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenCV
- **Version Control and Environment:** Git, GitHub

## Projects

- [🐙] **Cross-Task Capability Testing of SLMs**
- [🐙] **Sports Image Classification using a custom CNN model**
- [🐙] **Email Spam Classification using ML models**
- [🐙] **Credit Card Fraud Detection using ML models**

## References

1. **Nahid Hossain**
   *Assistant Professor and Program Coordinator*
   Department of Computer Science and Engineering
   United International University
   Email: nahid@cse.uiu.ac.bd
   *Relationship: Employer, Project Supervisor*

2. **Swakkhar Shatabda**
   *Professor*
   Department of Computer Science and Engineering
   BRAC University
   Email: swakkhar.shatabda@bracu.ac.bd
   *Relationship: Project Co-supervisor*

3. **Rafeed Rahman**
   *Senior Lecturer*
   Department of Computer Science and Engineering
   BRAC University
   Email: rafeed.rahman@bracu.ac.bd
   *Relationship: Thesis Supervisor*